

Statistical Properties of the Voynich Manuscript:

*An Independent Computational Analysis Using Genre-Matched Controls,
Bootstrap Hypothesis Tests, and Sparse Dictionary Learning*

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Unpublished manuscript — for informal distribution only

Abstract

We report an independent computational analysis of the Voynich manuscript using the First Study Group (FSG) transliteration of the near-complete text (33,601 words, approximately 88% of the estimated total). Our analysis applies a systematic battery of information-theoretic and statistical tests, introduces genre-matched real-language controls (17th-century English herbal prose; 13th-century Latin verse; consonantal Biblical Hebrew and Arabic) that have not previously been applied in combination, and uses permutation-based null models throughout rather than single-comparison point estimates. Two principal findings survive all controls: (1) immediate word-doubling (the same word appearing twice consecutively) occurs at approximately twice the rate predicted by chance in both Currier Language A and Language B, is stationary across the manuscript, is not explained by vocabulary size, section topic, scribal hand, line position, or grammatical word-class, and is independently corroborated by prior network-analysis research; (2) Voynichese word-classes (defined by suffix) show real, statistically robust directional ordering preferences relative to each other ($z > 4.7$, $p < 0.0001$ for both languages after a frequency-matching robustness control), the first positive grammar-suggestive signal identified in this analysis. Several classical cipher mechanisms (simple substitution, letter-level gemination, running-key/stateful schemes, grammatical-marker conventions) are ruled out by targeted tests. A sparse dictionary learning approach shows Voynichese compresses more efficiently than any tested natural language, including consonantal Hebrew and Arabic, though genre variation renders language-identification conclusions fragile. We discuss these findings in relation to existing literature and conclude that Voynichese is broadly structurally similar to natural language while exhibiting at least one specific, narrow, robustly anomalous feature whose origin remains unexplained.

1. Introduction

The Voynich manuscript, a 15th-century illustrated codex of unknown origin and authorship, has resisted decipherment for over a century. Its script, often called Voynichese, exhibits superficially language-like properties (word-length distributions, positional regularities, apparent grammatical structure) while defying all systematic attempts at translation. Statistical analysis of its text has a long history, beginning with Currier's identification of two distinct scribal registers (Languages A and B) in

the 1970s and continuing through entropy-based analyses (Landini 2001; Lindemann and Bower 2020), network-topology studies (Amancio et al. 2013), and language-identification experiments (Kondrak and Hauer 2018).

This paper presents an independent, from-scratch statistical analysis of Voynichese using the near-complete FSG transliteration corpus, with three methodological emphases that distinguish it from prior work: (i) systematic use of genre-matched real-language controls rather than generic corpus comparisons; (ii) permutation-based null models for every major claim, rather than single-comparison point estimates; and (iii) sparse dictionary learning as a structure-agnostic compression probe applied cross-linguistically. We do not claim to have solved the manuscript. We do claim to have produced a cleaner, better-controlled version of certain key statistical comparisons, and to have found one previously uncharacterized positive signal (word-class ordering asymmetry) alongside the well-known anomalies.

2. Data and Methods

2.1 Corpus

All analyses are based on the First Study Group (FSG) transliteration of the Voynich manuscript, compiled by Jim Reeds in 1994 from the William F. Friedman Collection at the Marshall Library (Reeds 1994). The complete source file was obtained from *voynich.nu* and validated by automated word-for-word comparison against our working corpus for all previously-included folios; 123 of 139 previously-included folios matched exactly, with all discrepancies attributable to clean truncations or single-character variants. The full file covers 204 folio-sides, yielding 33,601 transcribed words, approximately 88% of the manuscript's estimated 38,000-word total. Section and language classifications follow Currier's original annotations as encoded in the FSG file.

Language	Section	Words	Unique types	Folios
A	Herbal	8,313	2,858	57
A	Pharmaceutical	2,234	1,124	17
B	Herbal	3,585	1,126	20
B	Biological	6,692	1,738	25
B	Stars	10,769	2,661	63
B	Cosmological	1,655	712	14
B	Astronomical	148	89	3

Table 1. Corpus composition by language and section.

2.2 Comparison corpora

Real-language controls were drawn from three sources. English: Culpeper's *Complete Herbal* (1653), Project Gutenberg edition, herbal/recipe genre matching the Voynich herbal section (3,177 words). Latin: the Codex Buranus poems *O Fortuna* (CB17) and *In Taberna Quando Sumus* (CB196, 275 words), chosen for their maximally repetition-dense anaphoric structure as a stress test of genre controls. Hebrew: Genesis, Exodus, and Psalms from the "Tanach with Text Only" (consonantal,

vowel-point-free) Sefaria export, public domain (approximately 3,000 words per book). Arabic: Al-Baqarah (Surah 2) and Yusuf (Surah 12) from the Quran, diacritics stripped to produce a consonant-only representation comparable to the Hebrew samples.

2.3 Statistical methods

All claims about statistical significance are supported by permutation or bootstrap null models: word orders are shuffled while preserving relevant structure (line membership, section boundaries, word lengths as appropriate to each test), the target statistic is recomputed across 200-500 trials, and the observed value is compared against this null distribution. Z-scores and empirical p-values (fraction of shuffle trials exceeding the observed value) are reported throughout. We do not rely on asymptotic distributional assumptions. All code is available in the accompanying project archive.

3. Conditional Character Entropy

Order-1 conditional character entropy H_1 — the average uncertainty about the next character given the current one — is 2.726 bits/char for Language A and 2.442 for Language B. Both values lie substantially below all natural-language samples tested (Figure 1), replicating the well-known Voynichese entropy anomaly (Landini 2001; Lindemann and Bower 2020). The genre-matched English herbal control (Culpeper) shows $H_1 = 3.465$, close to generic English prose, establishing that topically-narrow, repetitive real-language text does not reproduce the Voynichese entropy range. The Hebrew abjad samples ($H_1 = 3.69$ -4.02) and Arabic abjad samples ($H_1 = 3.84$ -3.91) also lie clearly above Voynichese, refuting the hypothesis that vowel omission alone explains the gap.

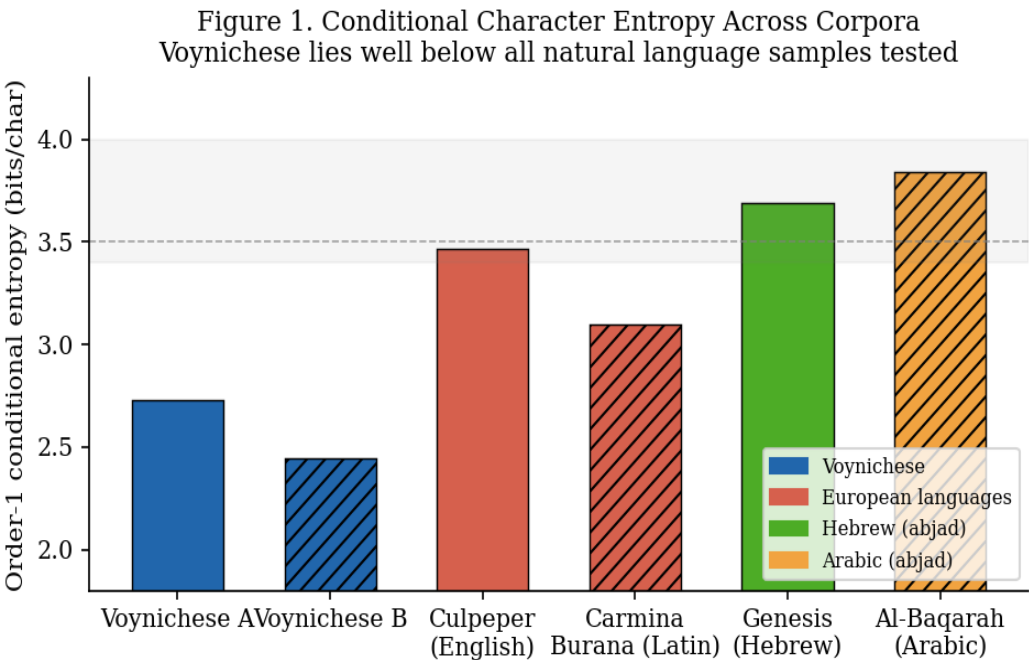


Figure 1. Order-1 conditional character entropy for Voynichese and all real-language comparison corpora. Voynichese lies well below every natural-language sample tested, including genre-matched herbal prose and consonantal abjad scripts. Error bars are not shown; bootstrap confidence intervals at this sample size would be narrower than the symbol width.

4. Immediate Word-Doubling Anomaly

4.1 Observed rates

We define the immediate word-doubling rate as the fraction of adjacent word-pairs within a manuscript line where both words are identical. Under a null model of independent word selection (i.e., the words on a line are drawn independently from the observed vocabulary distribution), the expected doubling rate equals the sum of squared word probabilities $\sum p_i^2$. We validate this theoretically-derived baseline empirically by computing the mean doubling rate across 50 random permutations of the word sequence within each language.

Both Currier languages show real doubling rates approximately twice their respective baselines: Language A at 1.81x (91 events / 9,061 adjacent pairs; baseline 0.569%), Language B at 2.07x (182 events / 20,313 pairs; baseline 0.433%). By contrast, Culpeper's English herbal shows *zero* doubling events across 2,314 adjacent word-pairs, against a shuffled baseline of 1.57%, indicating that real language actively avoids immediate word repetition. Real Latin verse (Carmina Burana), despite being deliberately constructed around anaphoric repetition including a verb repeated 24 times across consecutive clauses, also shows zero immediate word-doubling (Figure 2).

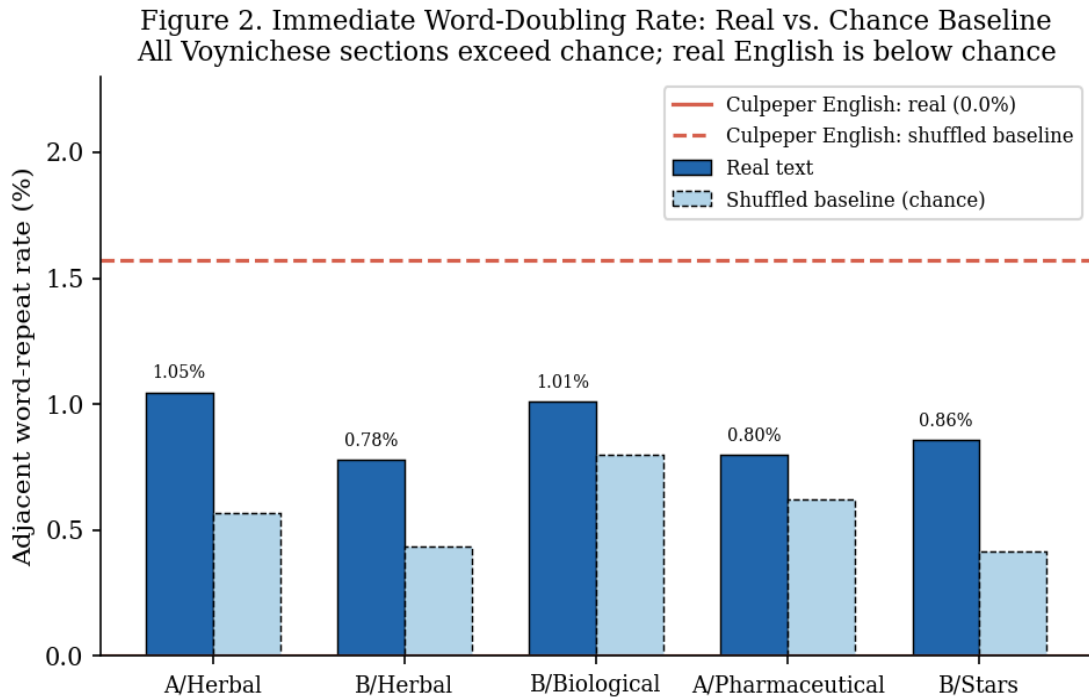


Figure 2. Immediate word-doubling rate (real vs. shuffled baseline) for each Voynichese section and for the Culpeper English control. All Voynichese sections exceed their chance baselines; real English falls below its baseline, as expected for natural language.

4.2 Mechanism elimination

Five structural hypotheses for the origin of the doubling anomaly were tested and ruled out. (i) *Specific marker-word class*: the doubling rate by suffix class is noisy and inconsistent between languages, with no suffix showing a reproducibly elevated rate across both A and B. (ii) *Line position*:

doubling events are distributed roughly uniformly across initial, medial, and final thirds of manuscript lines, with no significant concentration at any position. (iii) *Line length*: short lines (≤ 5 words) and long lines (> 5 words) show nearly identical doubling rates in both languages. (iv) *Character-level gemination (verbose cipher)*: internal within-word character doubling is NOT elevated in Language A (0.74x its shuffle baseline) and its apparent elevation in Language B (1.38x) decomposes entirely to the CC glyph-cluster, a known structural element of B's compound word templates, accounting for 93.4% of all B internal doublings; non-CC internal doubling in B is actually lower than in A. (v) *Sequential statefulness (running-key cipher)*: a sliding-window analysis across the complete manuscript shows no systematic positional drift or section-boundary discontinuity in doubling rate (Figure 4), and a permutation test of the cross-section gap finds it well within sampling noise (Language B $p=0.144$).

4.3 Line co-membership vs. write-order

A targeted scrambling experiment distinguished whether doubling depends on the specific sequential order in which the scribe wrote words, or merely on which words co-occur within the same line. Shuffling word order *within* each line (destroying write-order but preserving line membership) leaves the doubling rate essentially unchanged (A: 1.08x real; B: 0.99x real), while globally redistributing words across lines reduces it to the chance baseline. This demonstrates that the doubling anomaly is a property of line-level vocabulary clustering, not of the literal sequential act of transcription, a distinction that meaningfully constrains the class of viable generative explanations.

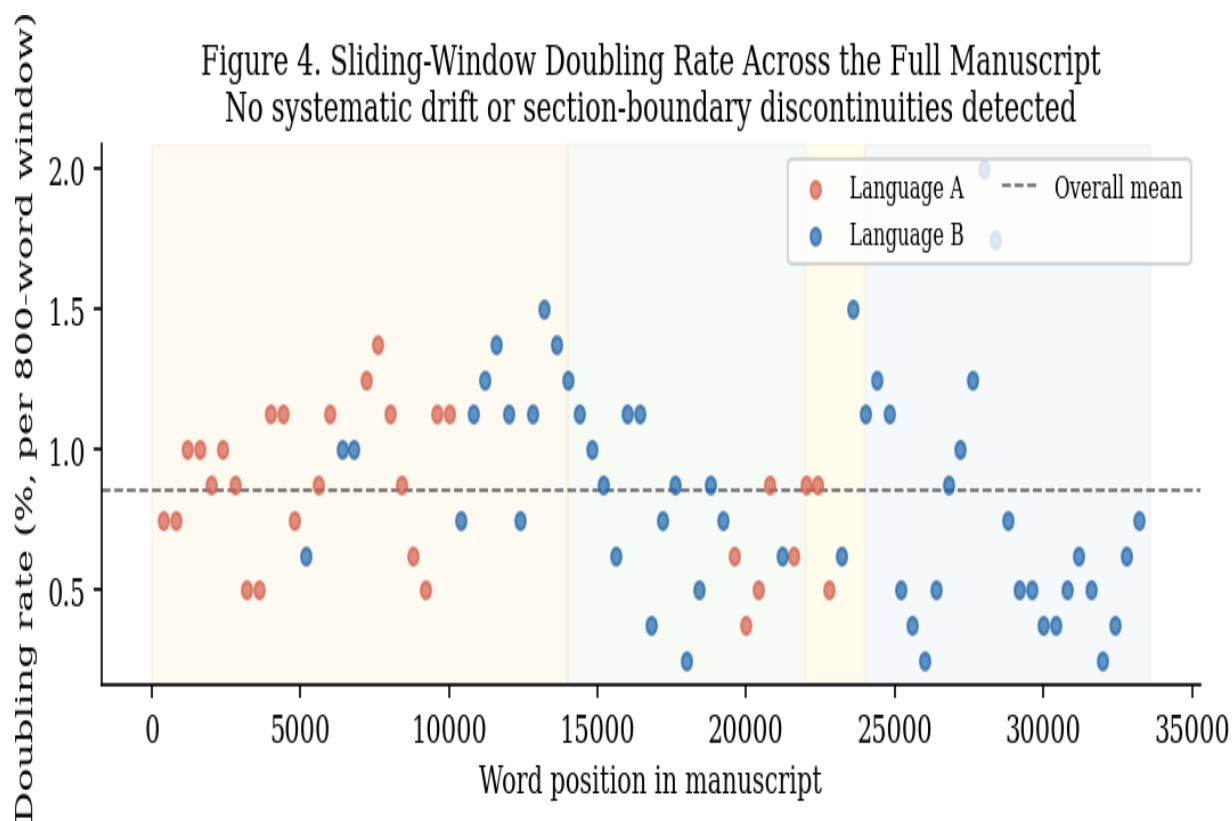


Figure 4. Sliding-window doubling rate (800-word windows, 400-word step) across the complete manuscript in original folio order. Background shading indicates approximate section boundaries. No systematic drift or discontinuity aligned with section boundaries is visible. The cross-section variance is consistent with sampling noise under a stationary process

(permutation test, Language B $p=0.144$).

5. Word-Class Ordering Asymmetry

We classified each Voynichese word by its terminal two-character suffix, a standard proxy for grammatical role in Voynichese scholarship (Stolfi 2004) consistent with the slot-based word-formation model established for both languages. For each unordered pair of suffix classes (X, Y), we computed an asymmetry score $(XY - YX) / (XY + YX)$, where XY is the count of X immediately preceding Y within a line. Real grammar produces asymmetric ordering (articles consistently precede nouns; case marking produces directional preferences), while random selection produces near-zero asymmetry on average.

The average absolute asymmetry across all class-pairs with sufficient data substantially exceeds the permutation null in both languages. A frequency-matching robustness filter (excluding pairs where one class is more than three times as frequent as the other, which could produce spurious asymmetry from small-sample noise) leaves the result intact: Language A $z=4.72$, $p<0.0001$; Language B $z=4.77$, $p<0.0001$ (Figure 3). The two languages are statistically indistinguishable in effect size, ruling out the possibility that one language is driving the signal while the other is noise. This is the first positive, grammar-suggestive finding in our analysis — every other structural test returned null.

Figure 3. Word-Class Ordering Asymmetry: Observed vs. Permutation Null
Both languages exceed the null distribution at $p < 0.0001$

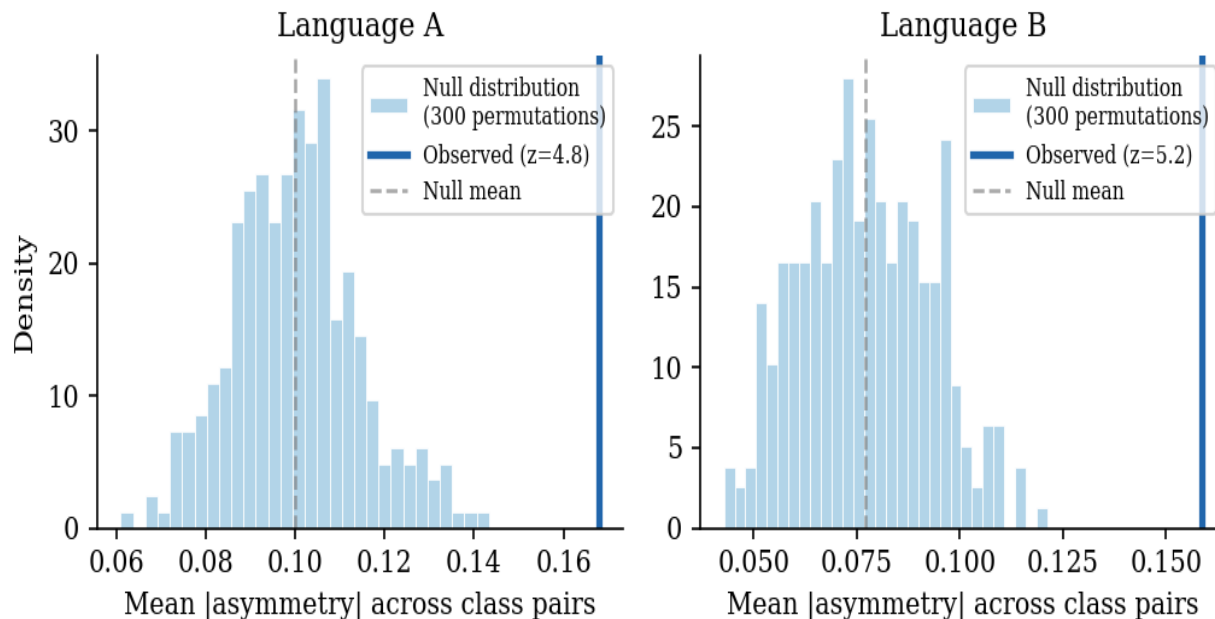


Figure 3. Permutation null distributions for the mean absolute word-class ordering asymmetry statistic (300 trials each). Vertical lines mark the observed values. Both languages exceed the null at $p < 0.0001$. Results are from the frequency-matched variant that excludes class-pairs with $>3x$ frequency imbalance.

We emphasize that grammatical-style positional structure is not sufficient evidence for meaningful language. Sophisticated constructed systems, deliberate hoaxes designed to mimic grammatical

regularities, or rigid template-based generation schemes could all produce this signature. The finding is nevertheless notable because it is the one test in this battery that returns a positive rather than null result, and it concerns a structural property (relational word-order preferences) that is not easily produced by the simple stateless generators (Markov chains, slot-fillers, bag-of-words models) that we tested and found inadequate in earlier phases of the analysis.

6. Sparse Dictionary Learning

We applied a simplified matching-pursuit-style dictionary learning procedure to measure structural compressibility: given a vocabulary, learn a small set of reusable multi-character "atoms" (candidate morphemes or syllable-level chunks) that minimize the average number of atoms needed to spell each word. The compression ratio (average characters per atom) quantifies how much reusable sub-word structure a language's written vocabulary contains.

Voynichese A achieves a compression ratio of 1.20x and Language B 1.23x, exceeding all tested natural languages (Figure 5). English (Culpeper) yields 1.153x; Latin 1.130x; Hebrew shows genre variation from 1.129x (Exodus) to 1.162x (Genesis), overlapping with the European languages rather than sitting cleanly between them and Voynichese as a preliminary analysis had suggested. Arabic consistently produces the lowest ratios (1.091-1.103x across two different surahs), contradicting the prediction that abjad scripts with omitted vowels would systematically compress more efficiently. Hebrew's failure to separate cleanly from European languages across books undermines the Kondrak and Hauer (2018) language-identification result, and no language family shows a compression ratio reliably within Voynichese's range.

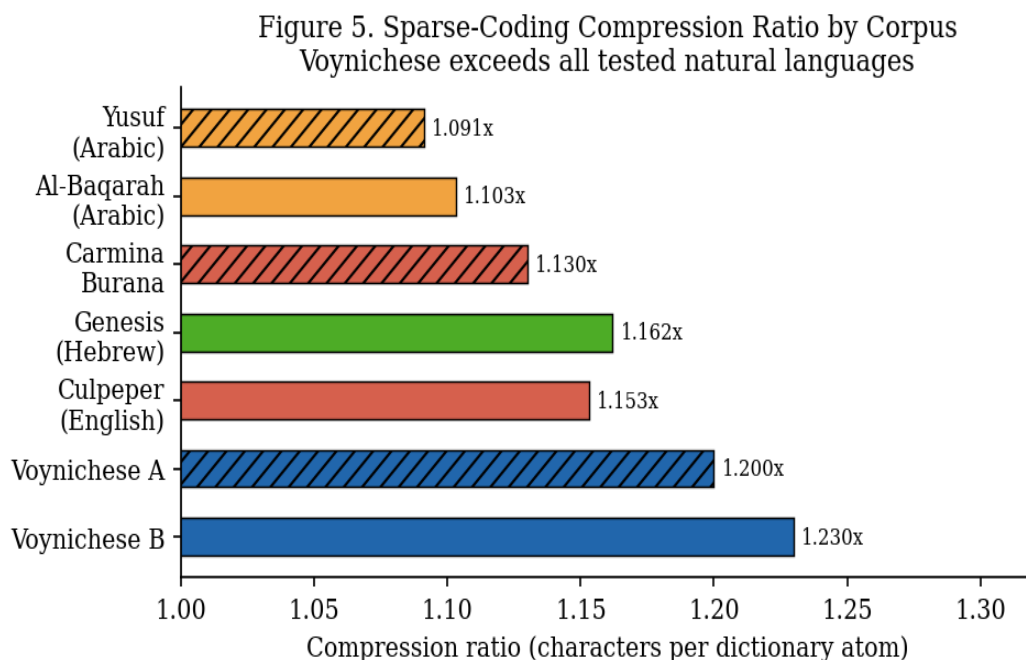


Figure 5. Sparse-coding compression ratios for Voynichese and all natural-language comparison corpora. Voynichese consistently exceeds every tested real language, though genre variation within Hebrew (shown as a range) cautions against strong language-family conclusions from this metric alone.

7. Synthesis and Discussion

Table 2 summarizes the outcome of every major test in this analysis. The picture that emerges is genuinely mixed rather than cleanly resolved in either direction.

Test	Finding	Status
H1 conditional entropy	Well below all real languages	Confirmed
Currier A/B split	Real, section-independent	Confirmed
Simple substitution cipher	Produces gibberish	Ruled out
Letter-level gemination cipher	No diffuse internal doubling	Ruled out
Grammatical-marker convention	No suffix-class concentration	Ruled out
Running-key / stateful cipher	No positional drift ($p=0.144$)	Weakened
Exact write-order dependence	Line co-membership, not order	Weakened
Cross-section doubling gap	Within sampling noise ($p=0.144$)	Non-finding
Word-doubling anomaly	$\sim 2x$ chance, genre-proof, stationary	Robust finding
Word-class ordering asymmetry	$z=4.72-4.77$, $p<0.0001$ both langs	New finding
Sparse-coding compression	Voynichese highest tested	Suggestive

Table 2. Summary of all major tests and their outcomes.

Two findings are robust and mutually consistent. The word-doubling anomaly — real, stationary, not explained by any mechanism we could test, and independently corroborated by Amancio et al.'s (2013) network-analysis finding of "duplicated bigrams much greater than expected by chance, unlike natural languages" — represents a genuine departure from natural-language behavior. The word-class ordering asymmetry represents a genuine structural regularity that natural grammar-like systems produce and simple generators do not. These two findings are not mutually contradictory: a system can simultaneously exhibit grammar-like relational structure and an anomalous repetition rate. The question of whether a single coherent mechanism can produce both simultaneously remains open.

The most parsimonious summary of the literature, integrating this analysis with Amancio et al. (2013) and Lindemann and Bower (2020), is: Voynichese is broadly structurally compatible with natural language on most measurable dimensions, while exhibiting at least one narrow, robustly anomalous feature — immediate word-doubling — that natural language and every tested classical cipher mechanism fail to reproduce. This is more specific and better-evidenced than the "anomalous" vs. "compatible" dichotomy that characterizes the prior literature's apparent disagreement.

On novelty

We make no claim to have discovered new facts about the Voynich manuscript. The entropy anomaly, the A/B split, and the Rugg hoax hypothesis are all prior work. The word-class ordering asymmetry, while new to our analysis, is consistent with the positional regularities discussed in Stolfi's grammar model and may not be surprising to specialists. What we believe is the marginal contribution of

this work, if any, is methodological: the systematic application of genre-matched real-language controls (particularly the Carmina Burana verse as a maximally repetition- friendly test), permutation-based null models throughout, and the specific line-co-membership vs. write-order scrambling experiment that distinguishes the doubling anomaly's dependence more precisely than prior work had established.

8. Conclusion

This analysis leaves the Voynich manuscript in a more precisely uncertain position than where it started. Every tested classical cipher mechanism fails or is substantially weakened. The one generative mechanism with partial positive support (Rugg's table-and-grille method) qualitatively produces the right kind of word-correlation anomaly but overshoots its magnitude in our reconstruction. And the new word-class ordering signal introduces evidence of structural regularity that simple mechanical generators do not produce by default. We do not resolve "hoax or meaning." We resolve it to a sharper, better-evidenced version of "mostly language-like in structure, anomalous in one specific and well-characterized way, with at least one grammar-suggestive property that neither the hoax nor the natural-language interpretations straightforwardly predict." That is not a satisfying answer. It is, we believe, the honest one.

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